

MTH3006 Portfolio Test

Syllabus: All content covered in the **first six weeks (practical classes and lectures)**. A **non-graphical calculator may be used**.

The test paper will include the following formulas:

1. Fourier transforms

$$\tilde{f}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt \quad \text{and} \quad f(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \tilde{f}(\omega) e^{i\omega t} d\omega$$

2. Fourier cosine transforms

$$\tilde{f}_c(\omega) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(t) \cos \omega t dt \quad \text{and} \quad f(t) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} \tilde{f}_c(\omega) \cos \omega t d\omega$$

3. Fourier sine transforms

$$\tilde{f}_s(\omega) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(t) \sin \omega t dt \quad \text{and} \quad f(t) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} \tilde{f}_s(\omega) \sin \omega t d\omega$$

4. Convolution for Fourier transforms

$$f * g = \int_{-\infty}^{\infty} f(x) g(z - x) dx$$

5. Convolution theorem for Fourier transforms

$$\mathcal{F}[f(x) * g(x)] = \sqrt{2\pi} \tilde{f}(k) \tilde{g}(k) \quad \mathcal{F}[f(x)g(x)] = \frac{1}{\sqrt{2\pi}} \tilde{f}(k) * \tilde{g}(k)$$

6. The step function and the delta function

$$H(t) = \begin{cases} 1 & \text{when } t > 0 \\ 0 & \text{when } t < 0 \end{cases} \quad \text{and} \quad \int_{-\infty}^{\infty} f(t) \delta(t - a) dt = f(a)$$

7. Laplace transforms

$$\tilde{f}(s) = \int_0^{\infty} f(t) e^{-st} dt$$

8. The shift and convolution theorems for Laplace transforms

$$\mathcal{L}[e^{-at} f(t)] = \tilde{f}(s + a) \quad \mathcal{L}\left[\int_0^t f(t - u) g(u) du\right] = \tilde{f}(s) \tilde{g}(s)$$

Table of Laplace transforms

Function	Laplace transform
e^{-at}	$1/(s + a)$
$\sin(\omega t)$	$\omega/(s^2 + \omega^2)$
$\cos(\omega t)$	$s/(s^2 + \omega^2)$
$t^n, n \geq 0$	$n!/s^{n+1}$
$t^p, p > -1$	$\Gamma(p + 1)/s^{p+1}$
$\sinh(at)$	$a/(s^2 - a^2)$
$\cosh(at)$	$s/(s^2 - a^2)$
y'	$s\tilde{y}(s) - y(0)$
y''	$s^2\tilde{y}(s) - sy(0) - y'(0)$

A non-graphical calculator may be used

1. Calculate the Fourier transform of

$$f(t) = \begin{cases} e^{-t} & \text{when } t \geq 0 \\ 0 & \text{when } t < 0 \end{cases}.$$

[10 marks]

2. Use the method of characteristics to solve

$$x \frac{\partial u(x, y)}{\partial x} + (1 + y) \frac{\partial u(x, y)}{\partial y} = 0$$

subject to the boundary condition

$$u = \left(1 + \frac{1}{x}\right)^2 \quad \text{on } y = x.$$

[15 marks]

3. Use separation of variables to solve

$$\frac{\partial u(x, t)}{\partial t} = \frac{\partial^2 u(x, t)}{\partial x^2}, \quad 0 < x < 2, \quad t > 0$$

subject to the boundary conditions $u(0, t) = u(2, t) = 0$ and $u(x, 0) = 3 \sin(2\pi x) + 2 \sin(4\pi x)$.

[15 marks]

4. Use Laplace transforms to evaluate the integrals

(a) $\int_0^\infty \sin 3t e^{-2t} dt$

(b) $\int_0^\infty (t - \cos(3t))e^{-t} dt$

[10 marks]

A non-graphical calculator may be used

1. Calculate the Fourier transform of

$$f(t) = \begin{cases} e^t & \text{when } -a < t < a \\ 0 & \text{otherwise} \end{cases}$$

where $a > 0$ is a real constant.

[10 marks]

2. Use partial fractions to find the inverse Laplace transform of

$$\tilde{f}(s) = \frac{s^2 - 15s + 41}{(s + 2)(s - 3)^2}$$

[15 marks]

3. Show that making the change of variables $\xi = y + 2x$ and $\eta = y - 2x$ transforms the differential equation

$$\frac{\partial^2 u}{\partial x^2} - 4 \frac{\partial^2 u}{\partial y^2} + \frac{\partial u}{\partial x} = 0$$

to

$$8 \frac{\partial^2 u}{\partial \xi \partial \eta} - \frac{\partial u}{\partial \xi} + \frac{\partial u}{\partial \eta} = 0.$$

[10 marks]

4. Use Laplace transforms to solve the differential equation

$$\frac{dx(t)}{dt} + 3x(t) = e^{-t}$$

given that $x(0) = 1$.

[15 marks]

A non-graphical calculator may be used

1. Calculate the Fourier transform of

$$f(t) = \begin{cases} t & \text{when } 0 \leq t \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

[10 marks]

2. Use the convolution theorem to calculate the inverse Laplace transform of

$$\tilde{F}(s) = \frac{1}{s^2(s^2 - 1)}.$$

[15 marks]

3. **Use Laplace transforms** to evaluate

(a) $\int_0^{\infty} e^{-3t} t^3 dt$

(b) $\int_0^{\infty} e^{-t} \cos at dt$, where a is a real, non – zero constant.

[10 marks]

4. Use the method of characteristics to solve

$$\frac{\partial u(x, y)}{\partial x} - x^2 \frac{\partial u(x, y)}{\partial y} = 2x^2 y$$

subject to the boundary condition $u = 1$ on $x = 0$.

[15 marks]